MILITARY SPECIFICATION

METER MODULATION ME-57( )/U

1. SCOPE

1.1 This specification covers one type of portable field instrument designated as Modulation Meter ME-57( )/U. The modulation meter is a compact lightweight portable field instrument capable of determining the accuracy of frequency deviation calibration of all types of fm signal generators and transmitters. The equipment measures to an accuracy of ± 5 percent, deviation over a range of 0 to 1 MHz at radio frequencies from 20 to 1,000 MHz, modulation frequencies from 50 to 70,000 Hz, with rf input levels as low as 5 mV. The overall accuracy is ± 10 percent. The equipment operates at either 115 or 230 Vac, ± 5 percent, 50 to 400 Hz. A transit case and cover is a part of the modulation meter.

2. APPLICABLE DOCUMENTS

2.1 Documents. The following documents of issue in effect at date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	Steel Strapping, Flat.
PPP-B-585	Box, Wood, Wirebound.
PPP-B-601	Box, Wood, Cleated-Plywood.
PPP-B-621	Box, Wood, Nailed and Lock-corner.
PPP-B-636	Box, Fiberboard.
PPP-B-640	Box, Fiberboard, Corrugated, Triple-Wall.
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.
PPP-P-291	Paperboard, Wrapping, Cushioning.
PPP-S-760	Strapping, Nonmetallic, (and Connectors).
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water-Resistant, (For Carton Sealing).
PPP-T-97	Tape; Pressure-Sensitive Adhesive, Filament Reinforced.

FSC 6625

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MIL-E-75	Electron, Tube Packing, Packaging, and Container Marking for General Specification for.
MIL-P-116	Preservation, Methods of.
MIL-P-11268	Parts, Materials and Processes used in Electronic Communication Equipment.
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Table for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-147	Palletized Unit Loads; 40 in. x 48 in. 4-Way Partial and 4-Way Pallets.
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462	Electromagnetic Interference Characteristics Measurement of.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-781	Reliability Test Exponential Distribution.
MIL-STD-810	Military Standard Environmental Test Methods for Aerospace and Ground Equipment.

DRAWINGS

ELECTRONICS COMMAND

SC-A-46439	List of Accessories for Package Tester.
SC-DL-320938	Meter Modulation ME-57( )/U.
SC-DL-399100	Transit Case Assembly.

Copies of specifications, standards and drawings required by contractors nection with specific procurement functions should be obtained from the ing activity, or as directed by the contracting officer. Both the title mber or symbol should be stipulated when requesting copies).

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3. REQUIREMENTS.

3.1 First article samples. The contractor shall furnish four (4) com- first article samples for approval provided the invitation for bid and contract requires first article samples.

3.2 Construction. The equipment shall be constructed in accordance the following drawings and the requirements specified herein, with Equip- Verification Review (EVR) in accordance with 4.10:

SC-DL-320938	Meter Modulation ME-57( )/U
SC-DL-399100	Transit Case Assembly

3.3 Parts, materials and processes-general. In addition to the require- s of this specification, the requirements of MIL-P-11268, including the ction of requirements therein, shall apply.

3.4 Finish, protective. The equipment shall be finished in accordance MIL-F-14072. This includes finish of hardware; such as screws, etc., necessary touch-up after mounting.

3.5 Marking. Marking shall conform to MIL-M-13231.

3.5.1 Visibility. Wherever practicable, parts and assemblies shall be ounted that their identification markings shall be readily visible with a mum disassembly of the equipment.

3.5.2 Serial numbers. Serial numbers are required on each nomenclatured name and type designation) item, except cords and cables.

3.6 Preconditioning.

3.6.1 Bounce preconditioning. Each equipment produced shall be subjected e bounce preconditioning of 4.6.1. After bounce preconditioning, the ment shall be inspected for loose or broken parts and controls. Bounce nditioning shall be performed prior to electrical preconditioning and A inspection.

3.6.2 Electrical preconditioning. Each equipment produced shall be cted to the electrical preconditioning of 4.6.2. Electrical precondition- hall be performed after bounce preconditioning and prior to Group A in- ion. The equipment shall be monitored for a 1 volt audio output in dance with 4.7.7 during period of test.

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3.7 Electrical.

3.7.1 Frequency range. The frequency range shall be continuous from 20 to 1,000 MHz in 5 bands, accurate to ±10 percent of indicated frequency, when tested in accordance with 4.7.1.

3.7.2 Deviation range and accuracy. The measurement of frequency deviation shall be continous from 0 to 1,000 kHz. The deviation accuracy at full scale shall be as follows, when tested in accordance with 4.7.2:

Deviation Range	Carrier: 20 to 100 MHz Modulation frequency: 50 to 20,000 Hz	Carrier: 100 to 1000 MHz Modulation frequency: 20,000 to 70,000 Hz
0-20 kHz	± .75 kHz dev.	± 1.5 kHz
0-50 kHz	± 2.0 kHz dev.	± 4.0 kHz
0-100 kHz	± 5.0 kHz dev.	± 10.0 kHz
0-300 kHz	± 15.0 kHz dev.	± 30.0 kHz
0-1000 kHz	± 100 kHz dev.	± 100 kHz

3.7.3 Frequency stability. The local oscillator following a 30 minute warmup period, shall not drift more than 200 Hz over a 10 second period for frequencies up to 400 MHz; for frequencies up to 1,000 MHz, the drift shall not exceed 1,000 Hz over a 10 second period. The equipment shall be tested in accordance with 4.7.3.

3.7.4 Amplitude modulation. The modulation meter shall be capable of operating in the presence of 50 percent ±5 percent amplitude modulation without deterioration for modulating frequencies of 60 Hz to 70 kHz, when tested in accordance with 4.7.4.

3.7.5 Carrier shift due to modulation. The modulation meter shall indicate carrier shift due to modulation from 10 to 250 kHz on a zero center meter indicator. The meter shall be calibrated in 10 kHz increments to 250 kHz and shall provide an accuracy of ±10 percent of its indicated value at the cardinal points, when tested in accordance with 4.7.5.

3.7.6 Input sensitivity. The modulation meter shall perform its specified function with input voltages ranging from .005 to 0.5 volt. The minimum sensitivity shall be .005 volt from 20 MHz to 1000 MHz. The equipment shall be tested in accordance with 4.7.6.

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3.7.7 Audio output. An audio output voltage of 1.0 volt shall correspond to an indicated frequency deviation of 40 kHz and shall be adjustable at the front panel within a range of ±10 percent. The 1.0 volt output shall remain stable to at least 1 percent for a period of not less than one-half hour. A linear relationship shall exist to within 1 dB between the detected audio voltage and frequency deviation down to 50 dB below 1 volt. The one volt shall remain constant to within 1 dB of the response at 1,000 Hz as the modulating frequency is varied from 100 Hz to 70 kHz ±5 percent, when tested in accordance with 4.7.7.

3.7.8 Distortion measurement. The harmonic distortion introduced by the modulation meter shall not exceed 3.0 percent up to 900 kHz frequency deviation at any of the modulation frequencies specified (50 Hz to 70 kHz), when tested in accordance with 4.7.8.

3.7.9 Noise and hum. The combined noise and hum output at the audio terminals with an unmodulated carrier input shall be at least 50 dB below one volt at 40 kHz deviation, with a flat weighing across a noise bandwidth of 50 Hz to 20 kHz, when tested in accordance with 4.7.9.

3.7.10 Frequency drift. The inherent frequency drift of the modulation meter shall not exceed ± 0.01 percent for any 10 minute period, when tested in accordance with 4.7.10.

3.7.11 Tuning dial accuracy. The accuracy of the modulation meter tuning dial shall be within 10 percent, when tested in accordance with 4.7.11.

3.7.12 AC power. The modulation meter shall operate from a 115 or 230 Vac, ±10 percent, 50 to 400 Hz, ±5 percent, single phase power source, when tested in accordance with 4.7.12.

3.7.13 Output impedance. The output impedance shall not vary more than ±10 percent over the frequency range of 100 Hz to 70 kHz. The output impedance across the two output binding posts shall be such that connection of distortion meters, noise meters and VTVM's which possess an input impedance of at least 10,000 ohms will not adversely affect the operation of the modulation meter, when tested in accordance with 4.7.13.

3.8 Service conditions. The modulation meter shall be constructed so that no fixed part shall become loose, cracked, flaked, pitted or blistered, no movable part or control shall be shifted in setting, position or adjustment, and no permanent degradation shall be caused in performance below that specified in the following service conditions.

3.8.1 Altitude. The modulation meter shall be subjected to the altitudes specified and shall meet the electrical requirements of 3.7.6 at times specified, when tested in accordance with 4.8.1.

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3.8.2 Low temperature. The modulation meter shall be subjected to the low temperature extremes of -4°F operating and -65°F storage and shall meet the electrical requirements of 3.7.6 at times specified, when tested in accordance with 4.8.2.

3.8.3 High temperature. The modulation meter shall be subjected to the high temperature extremes of 125°F operating and 160°F storage and shall meet the electrical requirements of 3.7.6 at times specified, when tested in accordance with 4.8.3.

3.8.4 Humidity. The modulation meter shall exhibit no physical damage of parts and materiel, such as corrosion, rust, blistering, swelling, deterioration, etc; or electrical degradation, and shall meet the electrical requirements of 3.7.6 at times specified, when tested in accordance with 4.8.4.

3.8.5 Immersion. The modulation meter transit case shall show no evidence of leakage, when tested in accordance with 4.8.5.

3.8.6 Vibration. The modulation meter shall show no evidence of breakage or loosening of parts, and shall meet the electrical requirements of 3.7.6, when tested in accordance with 4.8.6.

3.8.7 Bounce, loose cargo. The modulation meter shall show no evidence of physical damage except minor surface abrasion and shall meet the electrical requirements of 3.7.6, when tested in accordance with 4.8.7.

3.8.8 Shock, bench handling. The modulation meter shall show no evidence of physical damage except minor surface abrasion and shall meet the electrical requirements of 3.7.6, when tested in accordance with 4.8.8.

3.8.9 Fungus. The modulation meter shall show no evidence of viable fungus and shall meet the electrical requirements of 3.7.6, when tested in accordance with 4.8.9.

3.8.10 Electromagnetic interference. The modulation meter shall meet the interference and susceptibility requirements of Equipment Class Tactical Non-CE, Table A-I, of MIL-STD-461, when tested in accordance with 4.8.10.

3.9 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall meet Requirement 7 of MIL-STD-454. Inspection shall be in accordance with 4.9.

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3.10 Workmanship. The modulation meter shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454, and with the applicable portions of MIL-P-11268 as follows:

General requirements for plastic material and parts.
Wiring and cabling including:

Slack.
Protection.
Clearance.
Splicing.
Grounding, general.
Flux and cleaning agents for soldering.
Process for soldering.
Cleaning of equipment.
General process for securing of parts.
Process of welding.
Connectors, general.
Mechanical operation of controls.
Tropicalization of material.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the contract and specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (Does not include preparation for delivery) (see 4.3).

b. Inspection covered by subsidiary documents (see 4.4).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.5).

(2) Quality conformance inspection of preparation for delivery (see 4.12).

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4.3 First article inspection. This inspection shall be performed by the contractor and witnessed by a Government representative unless otherwise specified in the contract. It shall consist of Group A, B and C inspections (see Tables I, II and III respectively) on the four (4) samples of the completely assembled Modulation Meter ME-57( )/U. The first article samples shall be inspected in this order: Group A and B for all units; and designated units for specific Group C inspections. Designation of units to specific Group C inspections shall be determined by the Government representative. After completion of all Group C tests, conforming units must be reinspected and pass all Group A inspections.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection of equipment before preparation for delivery; with visual and mechanical inspection in accordance with Table I:

Item	Where required
Parts, materials and process-general	3.3
Finish	3.4
Marking	3.5
Workmanship	3.10

4.5 Quality conformance inspection of equipment before preparation for delivery. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedure and examine the contractor's inspection records. In addition, the Government-at its-discretion-may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. Each unit produced shall be inspected by the contractor for conformance to all Group A requirements and tests in Table I. Each lot shall be subject to contractor sampling inspection and Government verification, utilizing the procedures of MIL-STD-105 using the general inspection levels, and the AQL's indicated in Table I.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in any order which is satisfactory to the Government.

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TABLE I Group A inspection

Inspection	Req Para.	Test Para.	AQL Major	AQL Minor
Visual and mechanical				
Construction	3.2	4.11	1 dphu*	4 dphu
Parts, materials and processes general				
Finish, protective	3.3			
Marking	3.4			
Workmanship	3.5			
	3.10			
Electrical				
Frequency range	3.7.1	4.7.1	2.5 dphu for the group	**
Deviation range and accuracy	3.7.2	4.7.2		
Frequency stability	3.7.3	4.7.3		
Amplitude modulation	3.7.4	4.7.4		
Carrier shift due to modulation				
Input sensitivity	3.7.5	4.7.5		
Audio output	3.7.6	4.7.6		
Distortion measurement	3.7.7	4.7.7		
Noise and hum	3.7.8	4.7.8		
Frequency drift	3.7.9	4.7.9		
Tuning dial accuracy	3.7.10	4.7.10		
	3.7.11	4.7.11		

* Defects per hundred units.
** All electrical tests are major.

4.5.2 Group B inspection. This inspection, including contractor sampling and Government verification, shall conform to Table II and to the special procedure for small sample inspection of MIL-STD-105. Group B inspection shall be performed on inspection lots that have passed Group A inspection.

4.5.2.1 Order of inspection within Group B. Group B inspection shall be performed in any order which is satisfactory to the Government.

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TABLE II. Group B inspection

Inspection	Req. Para.	Test Para.	AQL
Interchangeability	3.9	4.9	6.5 dphu*
Electrical			
AC power	3.7.12	4.7.12	6.5 dphu for the group
Output impedance	3.7.13	4.7.13	

* Defects per hundred units.

4.5.3 Group C inspection. This inspection shall consist of the inspection specified in Table III and shall be performed on sample units that have been subjected to and met Group A and B inspection. Sample units shall be selected in accordance with 4.5.3.1.

4.5.3.1.1 Five (5) samples of complete modulation meter shall be selected at random, one (1) sample for each Group 1 through Group 5 in Table III. The samples selected shall be at the start of the contract from the first quality conformance inspection lot. The samples shall constitute Group C requirement for first 200 units produced.

4.5.3.1.2 Thereafter, two (2) samples of complete modulation meter shall be selected at random, one (1) sample for each Group 1 and Group 2 in Table III from every 200 units and fraction thereof produced.

4.5.3.2 Evaluation of samples submitted to individual Group C inspection. After referenced samples in 4.5.3.1 have been subjected to an individual Group C inspection test and prior to subjection to the next Group C test (unless otherwise specified in the individual test) the sample shall be subjected to and meet Input Sensitivity per 3.7.c and 4.7.6.

4.5.3.3 Noncompliance. The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Government Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Group A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C inspection results have shown that the corrective action was effective.

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4.5.3.4 Order of inspection within Group C. Group C inspection shall be performed in any order which is satisfactory to the Government.

TABLE III. Group C inspection

Inspection	Req. Para.	Test Para.
Group 1		
Altitude	3.8.1	4.8.1
Low temperature	3.8.2	4.8.2
High temperature	3.8.3	4.8.3
Humidity	3.8.4	4.8.4
Group 2		
Immersion	3.8.5	4.8.5
Vibration	3.8.6	4.8.6
Bounce, loose cargo	3.8.7	4.8.7
Shock, bench handling	3.8.8	4.8.8
Group 3		
Fungus	3.8.9	4.8.9
Group 4		
Electromagnetic interference	3.8.10	4.8.10
Group 5		
Construction	3.2	4.10

4.5.4 Reinspection of conforming Group B and C sample units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspection may be accepted on contract or purchase order provided all damage is repaired and the sample units are subjected to and pass Group A inspection.

4.5.5 Disposition of Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order provided units meet requirements of 4.5.4.

4.6 Preconditioning.

4.6.1 Bounce preconditioning. Each equipment produced shall be placed in its normal operating position on the table of a package tester of suitable capacity as made by the L.A.B. Corporation, Skaneateles, New York, with accessories as per Model 1000SC and SC-A-46439, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert

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a 1/32 inch thick strip of material under one corner or edge of the equipment under test to a distance of 3 inches as the equipment bounces. The equipment shall be subjected to this preconditioning for 1 minute and shall meet the requirements of 3.6.1.

4.6.2 Electrical preconditioning. Each equipment produced shall be placed in its normal operating position on a test table or in a test rack with room temperature at 77° ± 5°F, with specified operating voltage applied, from a suitable power source. With all controls in the normal operating position, the equipment shall be turned on and remain on for a continuous 24 hour period, and monitored. In case of parts failure, the equipment shall be repaired recalibrated, and the electrical preconditioning test repeated for the failed equipment. A failed item analysis report shall be submitted on each part failure with contents of report in accordance with 5.10.5 of MIL-STD-781. Each equipment shall meet the requirements of 3.6.2.

4.7 Electrical.

4.7.1 Frequency range. Utilizing the test set-up in figure 1, the modulation meter shall be tested and shall meet the requirements of 3.7.1 without hiatus. In the frequency range between 500 and 1,000 MHz, the signal generator (Hewlett-Packard 612A, or equal) may be used.

4.7.2 Deviation range and accuracy. Utilizing the test set-up in figure 2, set up a series of precise deviations using the panoramic indicator (Singer Metric SB-8B Type T-1000, or equal). The test results thus obtained shall meet the requirements of 3.7.2.

4.7.3 Frequency stability. Feed the output of the local oscillator of the modulation meter directly to the input of an electronic counter (Hewlett-Packard 5245L, or equal). After a 30 minute warm-up period, take frequency readings. Results thus obtained shall meet the requirements of 3.7.3.

4.7.4 Amplitude modulation. Feed the cw signal from an am signal generator (Hewlett-Packard 608E, or equal) to the modulation meter at a level sufficient to place the limiter indicator at the maximum point of the limiting range indicated on the meter scale. Applying 50 ± 5 percent modulation 100 Hz internal and 20 kHz external, there shall be no indication of carrier shift nor any indication on the deviation meter, other than residual incidental fm from the test am signal generator. The test results shall meet the requirements of 3.7.4. Note: Incidental fm from the test am generator may appear as a reading on the modulation meter. This reading shall be recorded.

4.7.5 Carrier shift due to modulation. Feed the output of the signal generator (Hewlett-Packard 608E, or equal) into the input of the modulation meter. Measure the frequency output of the generator with an electronic counter. The results of this test shall meet the requirements of 3.7.5.

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4.7.6 Input sensitivity. Utilizing the test setup in figure 1, and using a signal generator having a calibrated attenuator, feed in a signal until the limiting meter reads in the black area. Check the input voltage with an rf voltmeter (Hewlett Packard Model 3406, or equal). The modulation meter shall meet the requirements of 3.7.6.

4.7.7 Audio output. Utilizing the test setup in figure 4, the test fm signal generator modulation stability shall be verified before use in testing. A Hewlett Packard AC voltmeter Model 400 FL, or equal, shall be employed as the monitor for the 1 percent stability of the audio output. The modulation meter shall meet the requirements of 3.7.7.

4.7.8 Distortion measurement.

4.7.8.1 Prior to testing.

(a) Check that the total harmonic distortion of the signal generator plus the modulation meter, plus the distortion analyzer, does not exceed 3 percent.

(b) It is imperative that the measuring equipment be accurately calibrated to preclude an erroneous reading.

4.7.8.2 Testing.

(a) The modulation meter shall be tested to 1000 kHz deviation with the signal generator and the distortion analyzer (Hewlett Packard 334A, or equal). The modulation meter shall be tested above 400 MHz by using the second and third harmonics of the signal generator frequency, since the upper limit of the signal generator (Measurements Corp. H-4487, or equal) is 400MHz.

(b) It is permissable to adjust the modulation meter for minimum distortion, that is; a minor adjustment of the tuning dial to bring the test signal into the center of the modulation meter IF system.

(c) The modulation meter shall meet the requirements of 3.7.8.

4.7.9 Noise and hum. Utilizing the test set-up in figure 3, feed an unmodulated carrier into the modulation meter. Connect a band pass filter (Allison Labs. 420, or equal)(50 Hz to 20 kHz) across the audio output terminals. The reading after the filter shall meet the requirements of 3.7.9.

4.7.10 Frequency drift. The modulation meter shall be tested with a signal generator and a frequency counter for a 10 minute period, and without readjustment, shall meet the requirements of 3.7.10.

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4.7.11 Tuning dial accuracy. The modulation meter tuning dial calibration shall be checked from 50 to 400 MHz with a signal generator (Measurements Corp. H-4437, or equal), to 1,000 MHz with a signal generator (Hewlett-Packard 612A, or equal) on CW. The filled in sector of the limiting meter of the modulation meter shall be used as a tuning indicator, and shall meet the requirements of 3.7.11.

4.7.12 AC power. Connect the modulation meter to a power source capable of supplying ac voltages from 103.5 to 253 Vac at 50, 60, and 400 Hz, and maintain a minimum sensitivity of 5 millivolts under the following input line voltage conditions, and the equipment shall meet the requirements of 3.7.12:

a. 103.5 Vac at 50, 60 and 400 Hz.

b. 115 Vac at 50, 60, and 400 Hz.

c. 126.5 Vac at 50, 60, and 400 Hz.

d. 207 Vac at 50, 60, and 400 Hz.

e. 230 Vac at 50, 60, and 400 Hz.

f. 253 Vac at 50, 60, and 400 Hz.

4.7.13 Output impedance. Connect a Hewlett Packard Model 4800, or equal, impedance meter across the audio output terminals of the ME-57. The modulation meter shall meet the requirements of 3.7.13.

4.8 Service conditions. The modulation meter shall be tested in accordance with the following service conditions.

4.8.1 Altitude. The modulation meter in its normal operating configuration shall be subjected to the altitude test specified in Procedure I, Method 500 of MIL-STD-810, with altitude in Step 3 held for 1 hour, and shall meet the requirements of 3.8.1.

4.8.2 Low temperature. The modulation meter in its normal operating configuration shall be subjected to the low temperature test specified in Procedure I, Method 502 of MIL-STD-810, with temperatures in Steps 2 and 4 held for 24 hours each, and shall meet the requirements of 3.8.2.

4.8.3 High temperature. The modulation meter in its normal operating configuration shall be subjected to the high temperature test specified in Procedure I, Method 501 of MIL-STD-810, with temperature in Step 4 held for 24 hours, and shall meet the requirements of 3.8.3.

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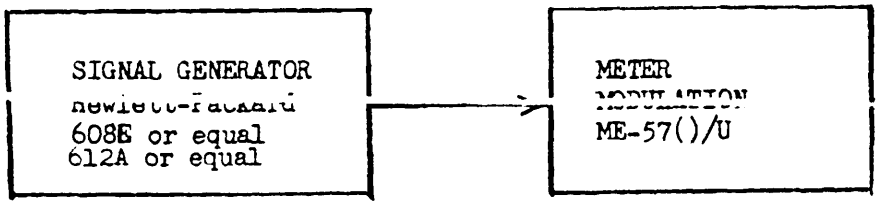


Figure 1

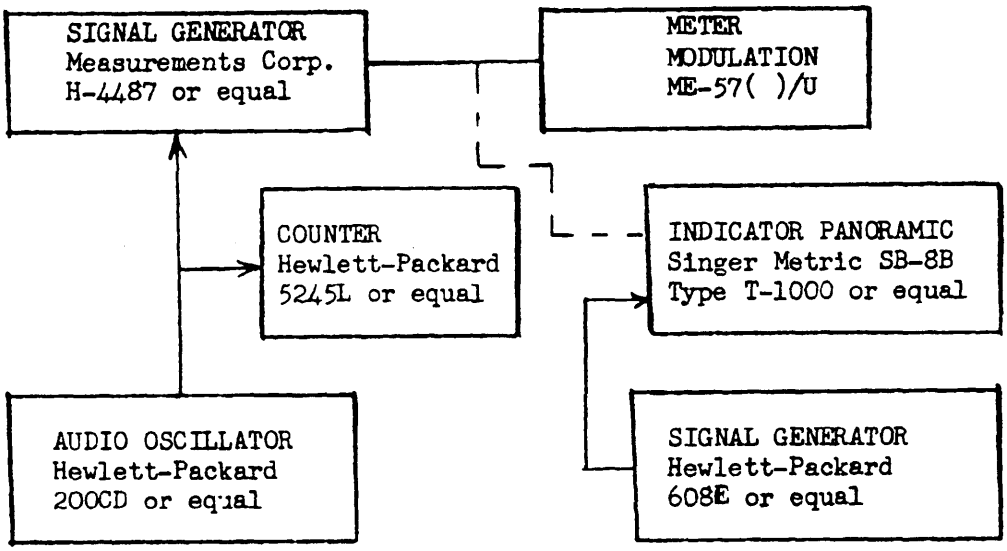


Figure 2

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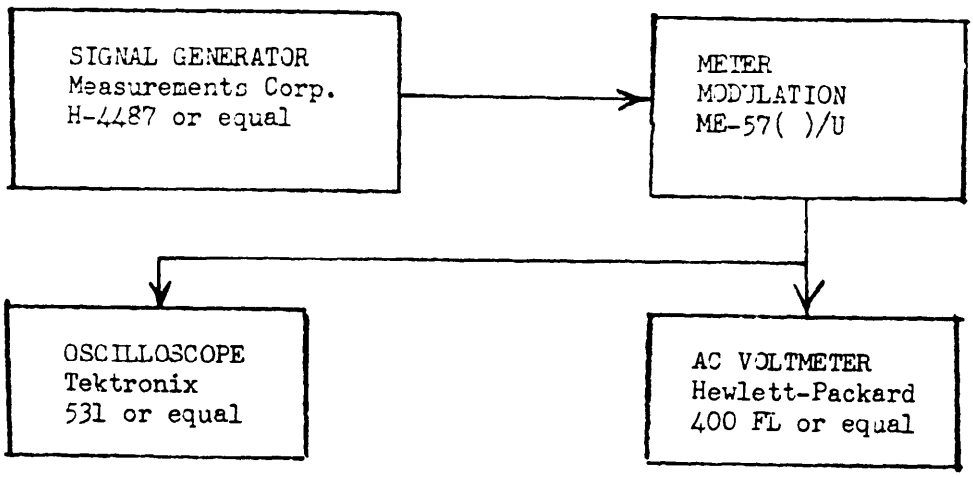


Figure 3

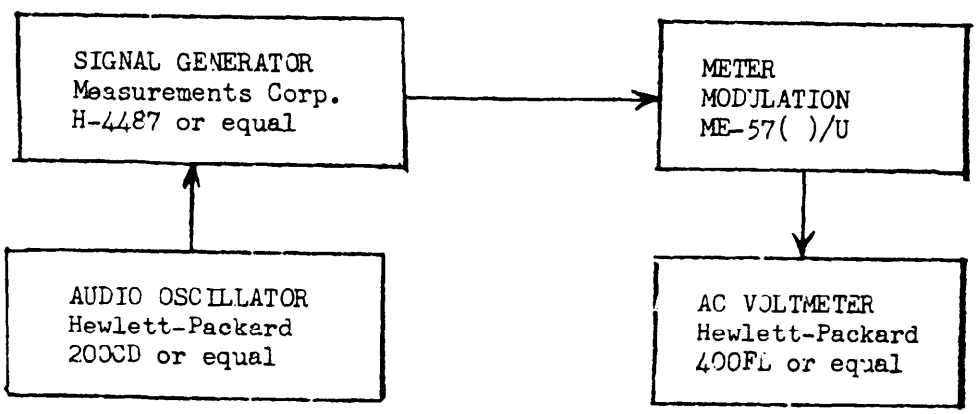


Figure 4

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4.8.4 Humidity. The modulation meter in its normal operating configuration shall be subjected to the humidity test specified in Procedure II, Method 507 of MIL-STD-810, and shall meet the requirements of 3.8.4.

4.8.5 Immersion. The modulation meter transit case, less the unit, shall be braced (for example: dummy front panel and chassis) and weighted to simulate the weight of the equipment, shall be subjected to the immersion test specified in Procedure I, Method 512 of MIL-STD-810, except the temperature of the water shall be 70° ± 5°F and the test item shall be 110° ±5°F, and shall meet the requirements of 3.8.5.

4.8.6 Vibration. The modulation meter shall be subjected to the vibration test specified in Part 1, Procedure IX, Method 514 of MIL-STD-810, and shall meet the requirement of 3.8.6.

4.8.7 Bounce, loose cargo. The modulation meter as prepared for field transportation shall be subjected to the loose cargo bounce test specified in Part 2, Procedure XI, Method 514 of MIL-STD-810, and shall meet the requirements of 3.8.7.

4.8.8 Shock, bench handling. The modulation meter shall be subjected to the bench handling shock test specified in Procedure V, Method 516 of MIL-STD-810, except 30 degree instead of 45 degree angle drop, and shall meet the requirements of 3.8.8.

4.8.9 Fungus. The modulation meter in its normal operating configuration shall be subjected to 28 days of the fungus test specified in Procedure I, Method 508 of MIL-STD-810 and shall meet the requirements of 3.8.9.

4.8.10 Electromagnetic interference. The modulation meter shall meet the requirements of 3.8.10. All test set-ups and procedures employed shall conform to the measurement techniques of MIL-STD-462.

4.9 Interchangeability. The dimensions listed below shall be measured to determine conformance to the physical interchangeability requirements of 3.9. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of case, cover, chassis and insertable assemblies, when such dimensions affect mating of parts.

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(b) Dimensions of oscillator assembly when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as case, covers, and mountings.

(d) Location of crystal test holders, gears and terminals. Gears shall have at least 80 percent mesh with the adjacent gear.

(e) Location of switches, sockets, and fasteners.

(f) Location of cover and fasteners on carrying case which must mate.

(g) Location of panel mounting screw holes on basic unit for mating.

4.10 Construction. The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the equipment on contract against the drawings cited for construction in 3.2. The EVR shall consist of the following.

4.10.1 An audit to establish that as built equipment (unit, assembly, module, or part) is in accordance with the cited end product drawings.

4.10.2 An audit to establish that as built configuration of a unit, assembly, module, or part meets the acceptance requirements (other than equipment specification performance) specified for each unit, assembly, module, or part. The audit shall record all acceptance test methods used together with resulting verification test data.

4.10.3 The contractor shall compile an EVR report containing the following information:

(a) Identification of unit, assembly, module, or part and details shall be in accordance with 4.10.1 and 4.10.2.

(b) Discrepancies noted.

(c) Corrective action taken.

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4.10.4 The contractor shall furnish the Government inspector a copy of the EVR report no later than 15 days prior to the submission of the first production unit or lot. The results of the EVR shall be subject to inspection by Government personnel at the time the first unit or production lot is offered for acceptance. Government inspection will be to the depth and extent necessary to demonstrate that the as built hardware is in accordance with the cited drawings. The contractor shall provide the following:

(a) Segregation of units, assemblies, modules, and parts to permit Government personnel access for detailed inspection.

(b) As required, responsible personnel from each functional department available for discussions in their respective areas.

(c) Adequate administrative support for the EVR.

4.11 Visual and mechanical. The equipment shall be subjected to a thorough visual and mechanical inspection. The defects shall be classified in accordance with MIL-STD-252.

4.12 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of Section 5.

5. PREPARATION FOR DELIVERY.

5.1 Preservation and packaging. Preservation and packaging and packing shall be level A or C as specified (see 6.2(b)).

5.1.1 Level A.

5.1.1.1 Cleaning. Each Meter, Modulation ME-57( )/U shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Each Meter, Modulation ME-57( )/U shall be dried in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Preservation application. None required.

5.1.1.4 Unit packaging. Unit packaging shall be in accordance with methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature. Each technical literature shall be packaged method 1C-1.

5.1.1.4.2 Electronic tubes. All tubes not securely clamped in place shall be removed and with spare tubes packaged individually in accordance with MIL-E-75.

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5.1.1.4.3 Spare fuses. Spare fuses, shall be packaged method III, as follows: Cushion each fuse by wrapping together in paperboard conforming to PPP-P-291, type 1, and secure the wrapping with gummed paper tape conforming to PPP-T-45, Grade A, type III.

5.1.1.4.4 Spare lamps. Spare lamps shall be individually packaged method III as follows: Cushion each lamp by wrapping in paperboard conforming to PPP-P-291, type 1, and secure wrapping with gummed paper tape conforming to PPP-T-45, Grade A, type III.

5.1.1.4.5 Consolidated package. Consolidate the items packaged as specified in 5.1.1.4.2 thru 5.1.1.4.4 within a close-fitting fiberboard box conforming to PPP-B-636, type CF, weather-resistant fill all voids with paperboard conforming to PPP-P-291, type II, to prevent movement. Box closure shall be as specified in the appendix of the box specification.

5.1.1.4.6 Meter, Modulation ME-57( )/U. Each Meter, Modulation ME-57( )/U shall be individually packaged method III as follows: Cushion each meter on all surfaces with cells or pads as both conforming to PPP-F-320, W5c, weather resistant, designed and fabricated to protect all projections and absorb the shock of impact in handling and transit. Fabricate a U shaped wood blocking frame the width and height of the fiberboard box to house and protect the consolidated spare parts package. Place the cushioned item with the wood blocking frame and spare parts consolidated package as specified in 5.1.1.4.5 together within a close-fitting fiberboard box conforming to PPP-B-636, W5c weather resistant. Place the technical literature packaged as specified in 5.1.1.4.1 in the box directly under the lid of the box. Box closure shall be as specified in the appendix of the box specification.

5.1.2 Level C. Meter, Modulation ME-57( )/U, shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. Packing shall be level A, B or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the containers or its contents.

5.2.1 Level A.

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5.2.1.1 Palletized load. A quantity of Meter, Modulation ME-57( )/U, packaged as specified in 5.1 shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.2 Less than palletized load. When quantities per destination are less than a pallet load, the item packaged as specified in 5.1, shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be placed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B.

5.2.2.1 Palletized load. A quantity of Meter, Modulation ME-57()/U packaged as specified in 5.1 shall be palletized as specified in 5.2.1.1.

5.2.2.2 Less than palletized load. When quantities per destination are less than a pallet load. A quantity of Meter, Modulation ME-57( )/U packaged as specified in 5.1, shall be placed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by nonmetallic strapping or pressure-sensitive filament tape banding conforming to PPP-S-760, type II, and PPP-T-97, type IV respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.3 Level C.

5.2.3.1 Palletized load. A quantity of Meter, Modulation ME-57( )/U packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1, except that the fiberboard caps shall be class domestic.

5.2.3.2 Less than palletized load. When quantities per destination are less than a pallet load, a quantity of Meter, Modulation ME-57( )/U packaged as specified in 5.1, shall be packed as specified in 5.2.2.2 except that the fiberboard boxes shall conform to PPP-B-640, and PPP-B-636, Class I and class domestic, respectively and reinforcing shall not be required for boxes conforming to PPP-B-636.

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5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES.

6.1 Intended use. Meter, Modulation ME-57( )/U is used for accurately calibrating frequency deviation of fm signal generators and fm transmitters. The modulation meter when used with external distortion analyzers, can be used for the measurement of modulation distortion. It can also be used for the measurement of fm signal generator and transmitter carrier shift due to modulation, for the measurement of fm spectrum distortion, for the measurement of fm noise and hum modulation, and in the measurement of spurious frequency modulation of am signal generators.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Levels of preservation, packaging and packing (see Section 5).

(c) When rough handling and functional tests are required.

(d) Quantity of first articles required.

(e) Place of final inspection.

(f) Technical literature required.

(g) Quantity of tools and running spare parts required.

(h) Marking for shipment and shipping containers.

6.3 Nomenclature and type designation. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. The parentheses in the type designation will be deleted or replaced by a letter identifying the particular design for example: SG-12F/U or AN/URM-73C.

6.4 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.5.3.3).

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6.5 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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